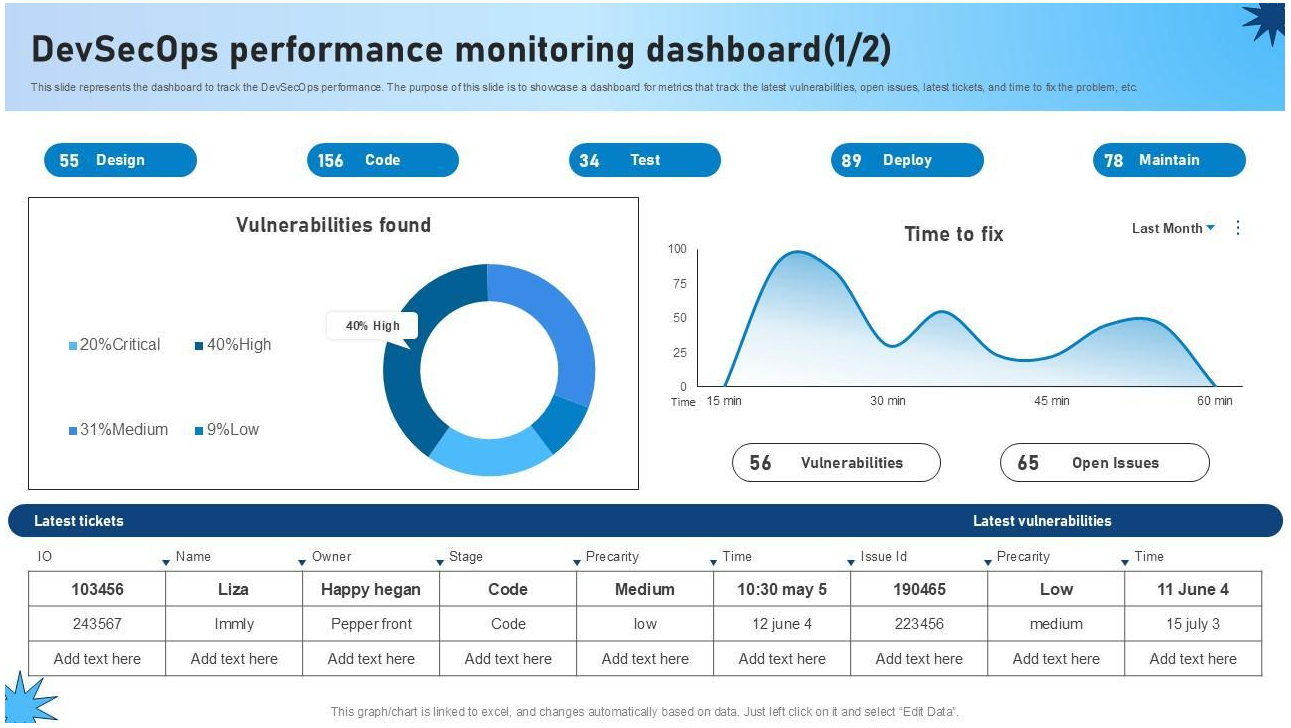
DevSecOps dashboards

Designing metrics for dashboards

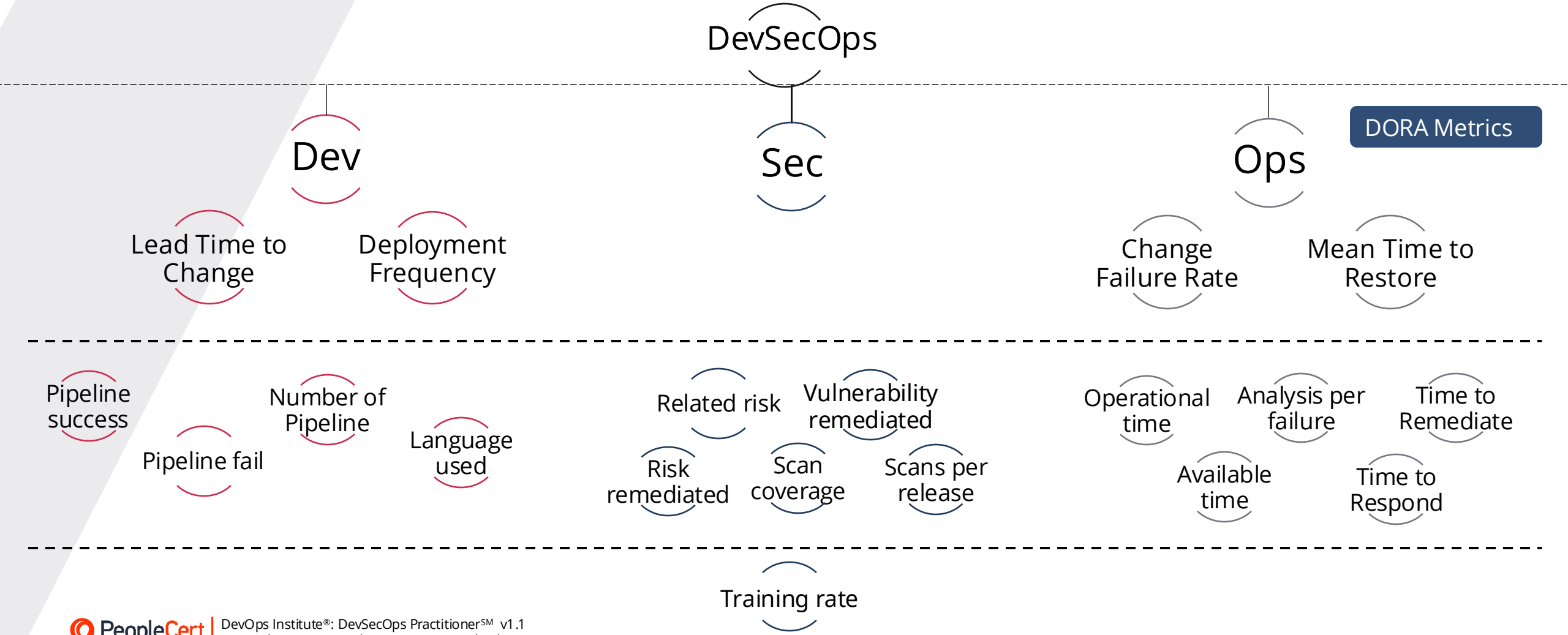
1. Define measurement goals.
2. Establish a baseline and targets.
3. Choose data sources and collection methods.
4. Implement data collection.
5. Validate and refine metrics.



Source: https://www.slidegeeks.com/business_powerpoint_diagrams/devsecops-performance-monitoring-dashboard-role-of-devsecops-ppt-presentation

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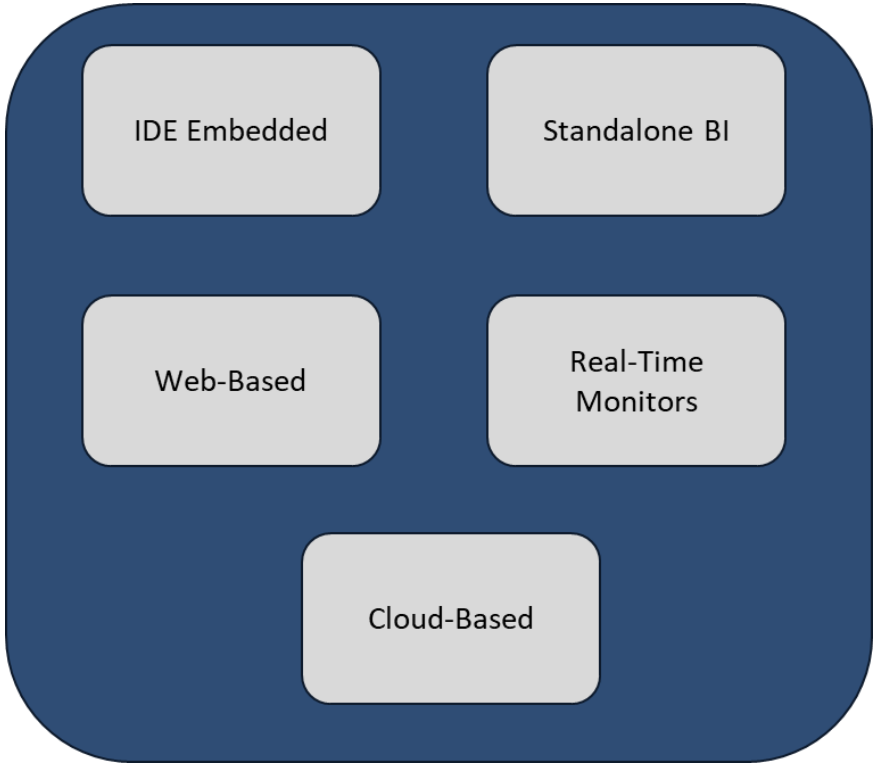
Building more analytics



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Architecting dashboards

- **Integrated Development Environment (IDE)-embedded dashboards:** developers see metrics in real time within their working environment, enhancing immediacy and relevance.
- **Standalone Business Intelligence (BI) tools:** tools such as Tableau, Power BI, or custom-built dashboard solutions pull data from various sources into a central dashboard.
- **Web-based dashboards:** allows highly customizable and accessible dashboards accessed via any web browser, across an organization.
- **Real-time monitoring dashboards:** integrate with databases and monitoring systems to provide live updates and alerts.
- **Cloud-based dashboards:** cloud platforms provide built-in tools to create dashboards that are scalable and accessible from anywhere.



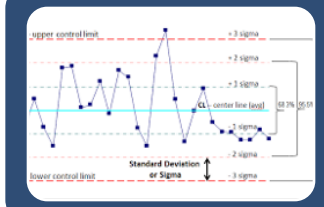
Source: *Continuous Testing, Quality, Security and Feedback* https://mybook.to/HtAAS8

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Control boundaries

Metrics require context

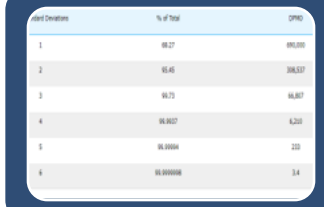
- **Six Sigma practice**
  - common cause variation
  - special cause variation.
- **Control boundaries** are another traditional PM concept with manufacturing.
- **Boundaries in DevOps**
  - flow metrics
  - vulnerability
  - load balancing
  - test coverage.



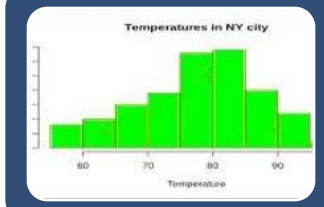
Control chart: measure standard deliverable.

Upper control limit/Lower control limit: mark the point beyond which a sample value is considered a special variation.

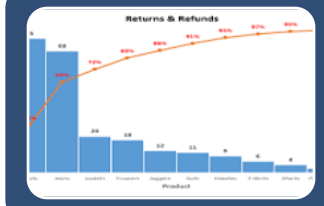
Centre line, where expected.



Sigma: standard deviation from the norm.



Histogram: divide into equal segments based on frequency.



Pareto chart: descending frequency magnitude.

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Case story: DevSecOps dashboard implementation at NASA



- 80% reduction in the development lifecycle
- 50% reduction in operations costs
- 30% reduction in infrastructure costs
- 2,000% increase in supported applications with no increase in staffing
- 16,000% increase in application traffic
- 90% of security vulnerabilities caught
- during the build process (shift left).

Metrics displayed:

- vulnerability detection rate
- compliance status
- build and deployment status
- incident response times.

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